Yujun Zhou

Personal Website | Google Scholar | YujunZhou | yzhou25@nd.edu

Summary

I am a fourth-year Ph.D. student in Computer Science and Engineering at the University of Notre Dame, advised by Prof. Xiangliang Zhang, working on large language models (LLMs).

My current research interests center around LLM reasoning, reinforcement learning, LLM Agents, and trustworthy LLMs. Broadly, I explore areas including LLM reasoning [4,5,8,9], LLM alignment [3, 6], trustworthy LLM evaluation and benchmarking [1], and LLM agents [2,6]. Please refer to my recent publications for additional details.

Previously, I was a research intern at Tencent AI Lab, working on LLM reasoning, especially math reasoning tasks [4, 8, 9].

I am currently a research intern at Microsoft, working on trustworthy LLM evaluation and realistic benchmark construction for LLM agents in complex enterprise-oriented scenarios.

Education

2023 - Present	PhD in Computer Science at **University of Notre Dame, USA**	(GPA: 4.00/4.0)
2021 - 2022	Master's Degree in Computer Science at **KAUST, Saudi Arabia**	(GPA: 3.83/4.0)
2017 - 2021	Bachelor's Degree in Computer Science at **UESTC, China**	(GPA: 3.92/4.0)

Selected Publications

[1] **Yujun Zhou**, Jingdong Yang, Kehan Guo, Pin-Yu Chen, Tian Gao, Werner Geyer, Nuno Moniz, Nitesh V Chawla, and Xiangliang Zhang. "LabSafety Bench: Benchmarking LLMs on Safety Issues in Scientific Labs". Published at ***Nature Machine Intelligence***.

[2] **Yujun Zhou**, Kehan Guo, Haomin Zhuang, Xiangqi Wang, Yue Huang, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Nuno Moniz, Nitesh V. Chawla, Xiangliang Zhang "Getting Better at Working With You: Compiling User Corrections into Runtime Enforcement for Coding Agents ". In *arxiv preprint*. arXiv:2606.13174.

[3] **Yujun Zhou***, Yue Huang*, Han Bao*, Kehan Guo, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Werner Geyer, Nuno Moniz, Nitesh V Chawla, Xiangliang Zhang "Capability-Oriented Training Induced Alignment Risk". In: ***ICML***. 2026.

[4] **Yujun Zhou***, Zhenwen Liang*, Haolin Liu, Wenhao Yu, Kishan Panaganti, Linfeng Song, Dian Yu, Xiangliang Zhang, Haitao Mi, Dong Yu. "Evolving Language Models without Labels: Majority Drives Selection, Novelty Promotes Variation". In *arxiv preprint*. arXiv:2509.15194.

[5] **Yujun Zhou***, Jiayi Ye*, Zipeng Ling*, Yufei Han, Yue Huang, Haomin Zhuang, Zhenwen Liang, Kehan Guo, Taicheng Guo, Xiangqi Wang, and Xiangliang Zhang. "Dissecting Logical Reasoning in LLMs: A Fine-Grained Evaluation and Supervision Study". In: ***EMNLP Findings***. 2025.

[6] **Yujun Zhou**, Yufei Han, Haomin Zhuang, Taicheng Guo, Kehan Guo, Zhenwen Liang, Hongyan Bao, and Xiangliang Zhang. "Defending Jailbreak Prompts via In-Context Adversarial Game". In: ***EMNLP***. 2024.

[7] **Yujun Zhou***, Yufei Han*, Haomin Zhuang, Hongyan Bao, and Xiangliang Zhang. "Attack-free Evaluating and Enhancing Adversarial Robustness on Categorical Data". In: ***ICML***. 2024.

[8] Zhenwen Liang*, **Yujun Zhou***, Sidi Lu, Xiangliang Zhang, Haitao Mi, Dong Yu. "Too Correct to Learn: Reinforcement Learning on Saturated Reasoning Data" In **ACL**. 2026.

[9] Zhenwen Liang*, Ruosen Li*, **Yujun Zhou***, Linfeng Song, Dian Yu, Xinya Du, Haitao Mi, Dong Yu. "Your Reasoning Model is Secretly a Reward Model - Optimization-Free Verification from

Experience". In **ACL**. 2026.

[10] Xiangqi Wang, Yue Huang, Yanbo Wang, Xiaonan Luo, Kehan Guo, **Yujun Zhou**, Xiangliang Zhang. "AdaReasoner: Adaptive Reasoning Enables More Flexible Thinking". In ***Neurips as a Spotlight***. 2025

Research Experiences

LLM Reasoning

- **On the Evolution of Language Models without Labels.** We developed EVOL-RL to solve "Entropy Collapse", a core failure mode in label-free RL where models lose reasoning diversity and fail to generalize. Our evolution-inspired framework balances consensus with a novelty reward, achieving state-of-the-art results by boosting pass@1 and pass@16 accuracy, preventing in-domain diversity collapse and improving out-of-domain generalization. (first-authored paper, arxiv)
- **Dissecting Logical Reasoning in LLMs: A Fine-Grained Evaluation and Supervision Study.** We proposed FineLogic, a fine-grained 3-D evaluation of LLM reasoning (accuracy, stepwise soundness, representation alignment) with a comparative evaluation of four supervision formats; we show that natural language supervision generalizes best and offer practical guidance for future directions to improve logical reasoning in LLMs with RL. (first-authored paper at EMNLP Findings 2025)

Trustworthy LLMs

- **Defending against Jailbreak Prompts via an In-Context Adversarial Game.** We propose to apply LLM agent learning to conduct adversarial training for LLMs without fine-tuning to defend against jailbreak attacks. (first-authored paper at EMNLP 2024)
- **Benchmarking LLMs on Safety Issues in Scientific Labs.** We propose LabSafety Bench, a specialized evaluation framework designed to assess the reliability and safety awareness of LLMs in laboratory environments. This extends the trustworthiness of LLMs to real-world safety-critical applications, such as lab safety, beyond the traditional focus areas. (first-author paper, published at Nature Machine Intelligence)
- **Capability-Oriented Training Induced Alignment Risk** We introduce capability-driven alignment risk, showing that RL-trained LLMs can unintentionally exploit training-time vulnerabilities and exhibit deceptive behaviors that mimic genuine performance gains. (first-author paper at ICML 2026)

LLM Agents

- **Making Coding Agents Follow User Corrections.** We show that coding agents can remember a user's correction yet still violate it in later sessions, and propose TRACE, a drop-in skill-layer pipeline that compiles each correction into a runtime check that must pass before task completion. TRACE reduces held-out preference violations from 100% to 37.6% while preserving task success. (first-authored paper, arXiv)

Working Experiences

Tencent Research Internship (2025.5-2025.12)

- **LLM Reasoning.** I was a full-time research intern at Tencent America LLC, focusing on reasoning in large language models (LLMs) as part of the AGI (Audio/Speech/Language & Multimodal) team.

  We proposed EVOL-RL that enables language models to evolve through label-free reinforcement learning, effectively addressing the issue of entropy collapse during training.

  Meanwhile, we also explored ways to expand the model's problem-solving boundaries, allowing it to tackle tasks that were previously entirely unsolvable.

Microsoft Research Internship (2026.5–2026.8)

- **Enterprise AI Benchmarking and Trustworthy LLM Evaluation.** I am working on building realistic enterprise-oriented benchmarks for evaluating LLM agents in complex workplace scenarios, with a focus on evidence-aware reasoning, reliable decision-making, and trustworthy behavior in long-horizon interactions.

Services

Workshop Organizer: RelSciFM Workshop, KDD 2026.
Invited Conference/Journal Reviewer: ICLR, NeurIPS, ARR, COLM, AAAI, KDD, CVPR, ICML